The photocathode that converts photons into photo electrons, the MCP that multiplies the photo-generated electrons, the phosphor screen that reconverts the electrons back into photons and the power supply that produces the electric field responsible for acceleration of electrons in different regions are all arranged in close proximity in an evacuated ceramic case.

Fig. 2: Microchannel plate

Efficiency with which the photocathode converts photons into electrons, also known as photocathode radiant sensitivity or quantum efficiency, depends on wavelength. A number of photocathode materials is in use. Of these, gallium arsenide and gallium arsenide phosphide crystals offer extremely high sensitivity. Photo electrons are accelerated by an electric field produced by a high voltage applied between the photocathode and the MCP input surface.

The MCP is a thin glass disk, about 0.5mm thick, consisting of an array of millions of tilted glass channels, each five to six microns in diameter, bundled in parallel (Fig. 2). A single-stage MCP used in second-generation intensifier tubes provides electron multiplication of about 10^3. Two- and three-stage MCPs produce a gain of 10^5 and greater than 10^6. First-generation tubes did not use an MCP.

The number of stages to be used in the MCP depends on the required value of gain. Strip current that flows through the MCP decides the dynamic range or linearity of the image intensifier tube. A low-resistance MCP causing a large strip current to flow through the MCP is desirable for achieving high linearity.

The phosphor screen reconverts the impinging electrons back to photons. Commonly-used phosphor types include P24, P43, P46 and P47. Phosphor screens are characterised by peak emission wavelength, decay time, power efficiency and emission colour.

Phosphor screen decay time is one of the most important parameters to be considered while selecting a suitable phosphor type. The ideal phosphor type is one whose decay time matches the readout method and whose spectral emission matches the readout sensitivity. When used with a linear image sensor or high-speed CCD, a short decay time is recommended for the phosphor screen, to avoid the appearance of an after-image in the next frame. On the other hand, a short decay time that minimises flicker is recommended for night time viewing and surveillance applications.

Fig. 3: Different package configurations of night vision devices

Output window material is selected to match the readout method. Different output window types include borosilicate glass, fibre-optic plate and twisted fibre-optics.

A borosilicate glass window is used for relay lens readout. In this case, the relay lens is focused on the phosphor screen.

Fibre-optic output plate—a standard output window—is ideal for direct coupling to a CCD with fibre-optic plate input window. It consists of millions to hundreds of millions glass fibres bundled in parallel. A fibre-optic plate can transmit an optical image from one surface to another without causing any distortion. Diameter of the glass fibre matches the channel diameter of the MCP.

Twisted fibre optics as output window are used for night time viewing applications. Twisted fibre optics reduce eyepiece length, thereby making the night vision device more compact.

Image intensifier tube-based night vision devices are made in a variety of package configurations (Fig. 3), according to the requirements of specific application scenarios. These mainly include night vision monoculars, binoculars, weapon sights and goggles.

Operational modes. There are two common modes of operation of an image intensifier tube: gated and photon counting. In the gated mode of operation, the intensified image can be gated to open or close the optical shutter by varying the potential difference between the photocathode and the inside surface of the MCP, thereby either allowing or disallowing the formation of an intensified image.

In gate-on mode, potential of the photocathode is lower than that of the MCP. As a result, photoelectrons are attracted towards the MCP, to be subsequently multiplied and hit the phosphor screen, producing an intensified image.

In gate-off mode, potential of the inside surface of the MCP is less than that of the photocathode so that photo electrons can revert back to the photocathode. Therefore no intensified image is seen on the phosphor screen.

In practice, MCP potential is fixed, and the intensifier tube is turned on by applying a negative polarity pulse of about 200 volts to the photocathode. Gated operation is very effective in analysing high-speed optical phenomenon.

Image intensifier tubes using a three-stage MCP have much higher sensitivity than those employing a single-stage MCP. This is important when it comes to operating at extremely low-light levels. When the light level is as low as 10^{-4} lux, a three-stage MCP helps in producing an image of acceptable quality. However, when the light level falls below 10^{-5} lux, incident photons are separated in time and space.

It is no longer possible to capture an image with a gradation. At extremely low-light levels, when only a few light spots appear on the phosphor screen per second, a good-quality image can be obtained by detecting each spot and its position, and integrating these into an image storage unit. Brightness distribution in photon counting mode is given by the difference in the number of photons at each position.

Different generations of image intensifiers. Night vision technology has undergone substantial changes in the last 40 years. These changes have led to gigantic improvements in performance standards of night vision devices. Each substantial change in technology is associated with a generation. We have seen several generations of night vision devices based on image intensifier tube technology. Beginning with generation-0, we are currently on generation-4.